

Creating Microservices for account and loan

Account.java:

```
package com.cognizant.account.model;

public class Account {

    private String number;
    private String type;
    private double balance;

    public Account() {}

    public Account(String number, String type, double balance) {
        this.number = number;
        this.type = type;
        this.balance = balance;
    }

    public String getNumber() {
        return number;
    }

    public void setNumber(String number) {
        this.number = number;
    }

    public String getType() {
        return type;
    }

    public void setType(String type) {
        this.type = type;
    }

    public double getBalance() {
        return balance;
    }

    public void setBalance(double balance) {
        this.balance = balance;
    }
}
```

AccountController.java:

```
package com.cognizant.account.controller;

import com.cognizant.account.model.Account;
import org.springframework.web.bind.annotation.*;

@RestController
public class AccountController {

    @GetMapping("/accounts/{number}")
    public Account getAccount(@PathVariable String number) {
        return new Account(number, "Savings", 234343);
    }
}
```

Loan.java:

```
package com.cognizant.loan.model;

public class Loan {

    private String number;
    private String type;
    private double loan;
    private double emi;
    private int tenure;

    public Loan() {}

    public Loan(String number, String type, double loan, double emi, int tenure) {
        this.number = number;
        this.type = type;
        this.loan = loan;
        this.emi = emi;
        this.tenure = tenure;
    }

    public String getNumber() {
        return number;
    }

    public String getType() {
        return type;
    }

    public double getLoan() {
        return loan;
    }
}
```

```

public double getEmi() {
    return emi;
}

public int getTenure() {
    return tenure;
}
}



## LoanController.java:



package com.cognizant.loan.controller;

import com.cognizant.loan.model.Loan;
import org.springframework.web.bind.annotation.*;

@RestController
public class LoanController {

    @GetMapping("/loans/{number}")
    public Loan getLoan(@PathVariable String number) {
        return new Loan(number, "Car", 400000, 3258, 18);
    }
}

```

Output:

```

adityahargunani@Adityas-MacBook account % mvn spring-boot:run

Spring
:: Spring Boot :: (v4.1.0)

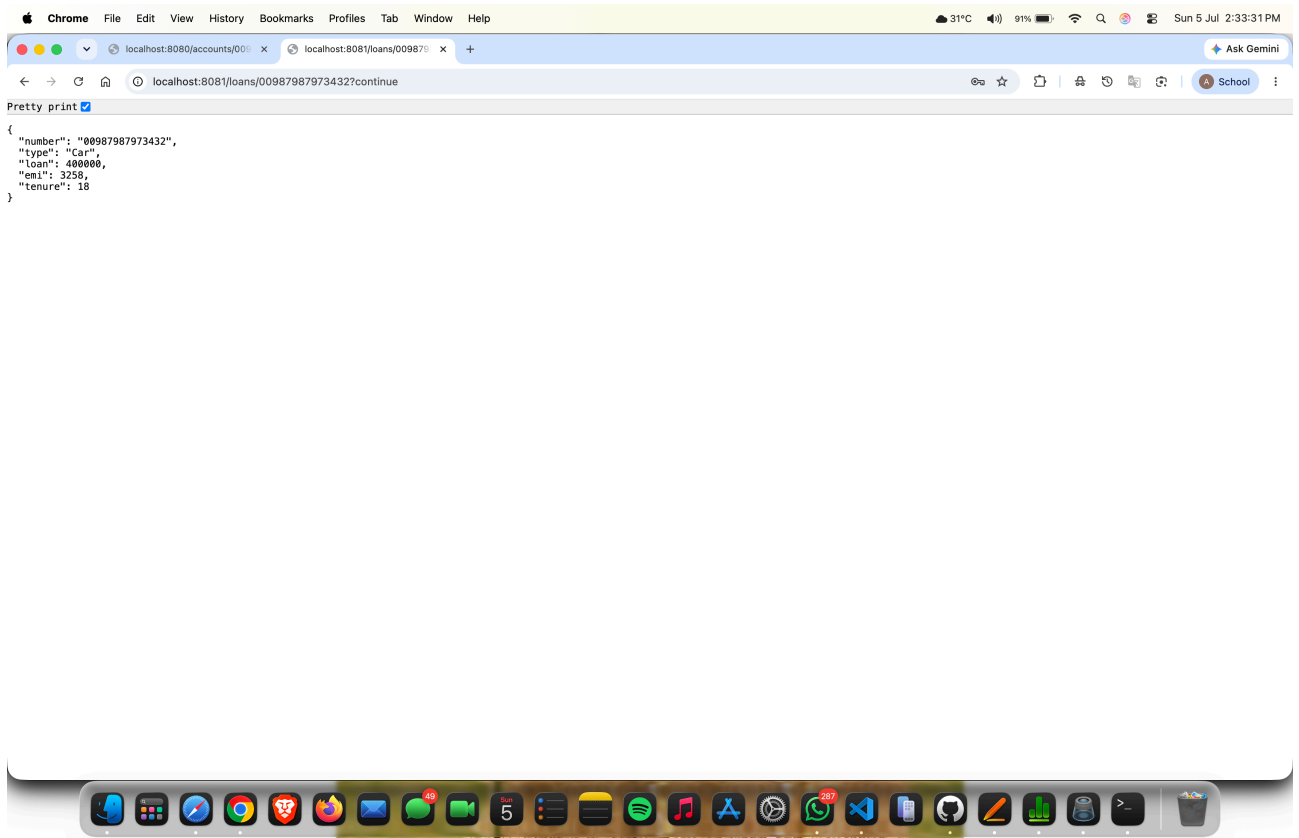
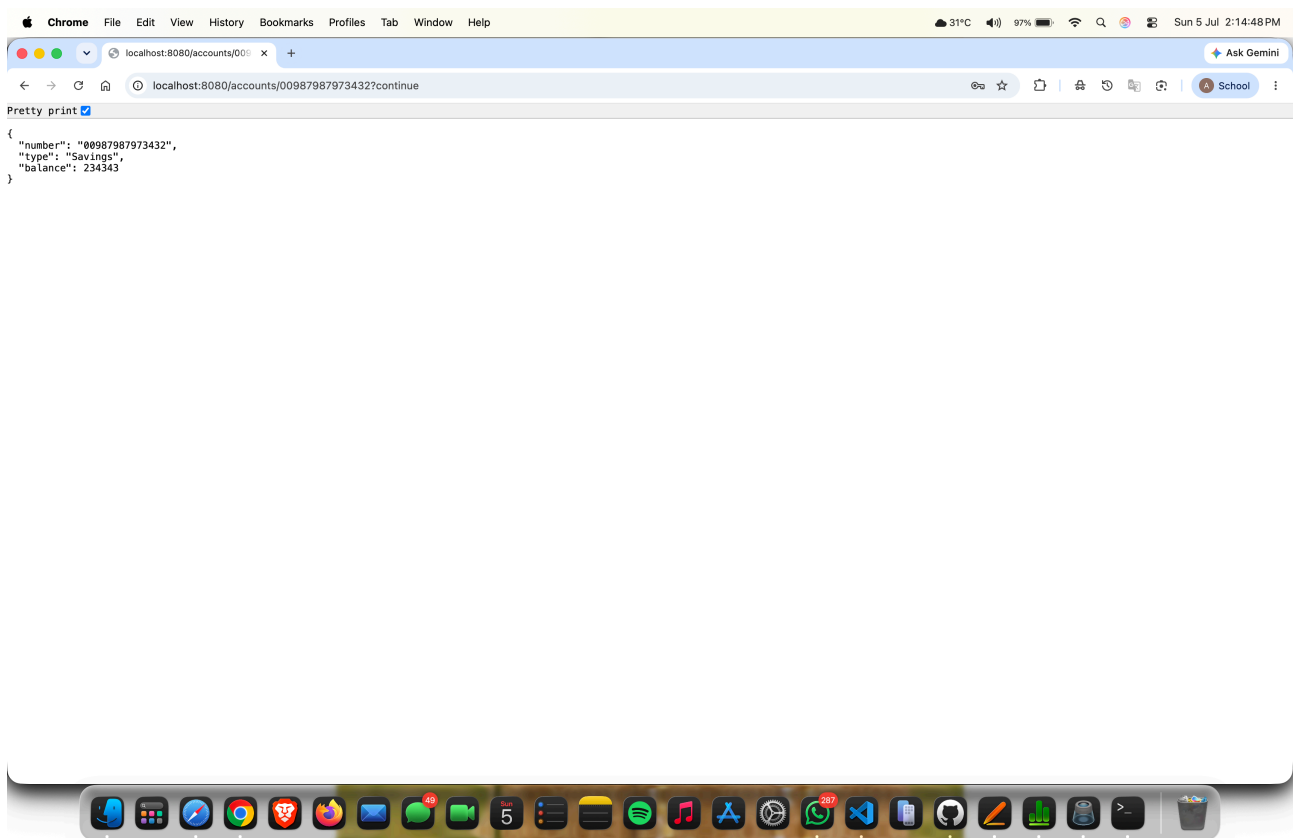
2026-07-05T14:13:31.970+05:30 INFO 35301 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Starting AccountApplication using Java 26.0.1 with PID 35301 (/Users/adi
tyahargunani/Downloads/aditya/account/target/classes started by adityahargunani in /Users/adityahargunani/Downloads/aditya/account)
2026-07-05T14:13:31.971+05:30 INFO 35301 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : No active profile set, falling back to 1 default profile: "default"
2026-07-05T14:13:31.988+05:30 INFO 35301 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.add-properties'
to 'false' to disable
2026-07-05T14:13:31.988+05:30 INFO 35301 --- [account] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : For additional web related logging consider setting the 'logging.level.w
eb' property to 'DEBUG'
2026-07-05T14:13:32.249+05:30 INFO 35301 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-07-05T14:13:32.254+05:30 INFO 35301 --- [account] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-07-05T14:13:32.254+05:30 INFO 35301 --- [account] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/11.0.22]
2026-07-05T14:13:32.263+05:30 INFO 35301 --- [account] [ restartedMain] b.w.c.s.WebApplicationContextInitializer : Root WebApplicationContext: initialization completed in 275 ms
2026-07-05T14:13:32.355+05:30 WARN 35301 --- [account] [ restartedMain] .s.a.UserDetailsServiceAutoConfiguration :

Using generated security password: a8aa5123-8fb5-4640-83d9-ddbc9e0d6bdb

This generated password is for development use only. Your security configuration must be updated before running your application in production.

2026-07-05T14:13:32.366+05:30 INFO 35301 --- [account] [ restartedMain] rs.InitializeUserDetailsServiceConfigurer : Global AuthenticationManager configured with UserDetailsService bean wit
h name inMemoryUserDetailsService
2026-07-05T14:13:32.418+05:30 INFO 35301 --- [account] [ restartedMain] o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-07-05T14:13:32.420+05:30 INFO 35301 --- [account] [ restartedMain] c.cognizant.account.AccountApplication : Started AccountApplication in 0.571 seconds (process running for 0.707)
2026-07-05T14:13:42.845+05:30 INFO 35301 --- [nio-8080-exec-1] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring DispatcherServlet 'dispatcherServlet'
2026-07-05T14:13:42.846+05:30 INFO 35301 --- [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Initializing Servlet 'dispatcherServlet'
2026-07-05T14:13:42.846+05:30 INFO 35301 --- [nio-8080-exec-1] o.s.web.servlet.DispatcherServlet : Completed initialization in 1 ms

```



```
● adityahargunani@Adityas-MacBook account % curl -u user:b91c2750-a177-4a27-941f-6fcc3328bffd http://localhost:8080/accounts/00987987973432
{"number":"00987987973432","type":"Savings","balance":234343.0}
○ adityahargunani@Adityas-MacBook account %
```

```
● adityahargunani@Adityas-MacBook loan % curl -u user:720b91a7-96d2-4a2f-99d7-abb615050b2d http://localhost:8081/loans/00987987973432
{"number":"00987987973432","type":"Car","loan":400000.0,"emi":3258.0,"tenure":18}
○ adityahargunani@Adityas-MacBook loan %
```